

Week 10!?!

- Go over code beta test
- The “missing link” in our course

Coding quiz

100%

High Score

60%

Low Score

89%

Mean Score

93%

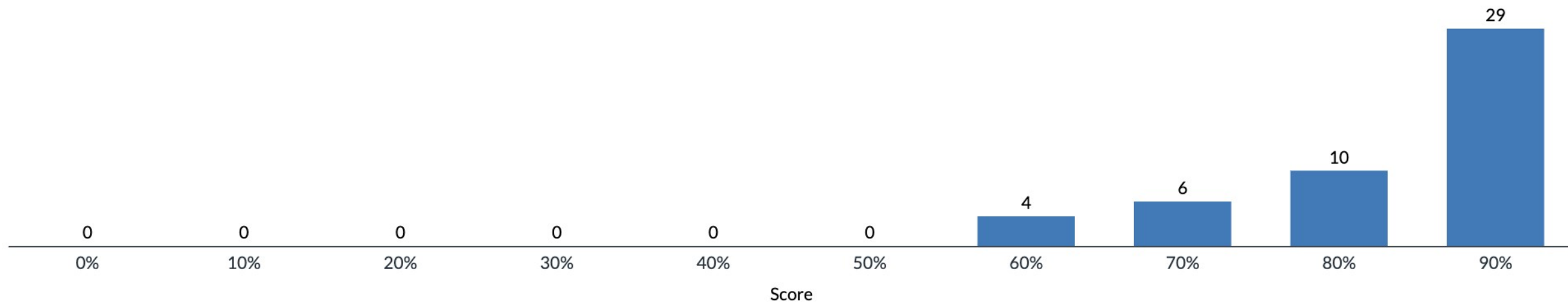
Median Score

00:16:05

Mean Elapsed
Time



Score Distribution



When preparing training data for the DataLoader, we used something like ``torch.stack([tokenize(n) for n in train_names])``. What does ``torch.stack`` do here?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Combines a list of individual tensors into a single batched tensor with a new first dimension	44	90%
✗	Concatenates all token tensors end-to-end into one long 1D tensor	3	6%
✗	Pads each tensor to the same length	2	4%
✗	Converts strings to tensors	0	0%

In the In-Context Learning (ICL) method, we included solved examples in the prompt. How were these examples structured in the messages list?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Examples were given as a pair of messages with questions from "role":"user" and answers from "role":"assistant"	38	78%
✗	Examples were all placed in a single markdown formatted table.	0	0%
✗	Examples were appended to the system prompt with instructions to answer in a manner similar to the provided examples.	11	22%

Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?

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>2k cites

“other aspects of the demonstrations are the key drivers of end task performance:”

(1) the label space

(2) the distribution of the input text, and

(3) the overall **format of the sequence**

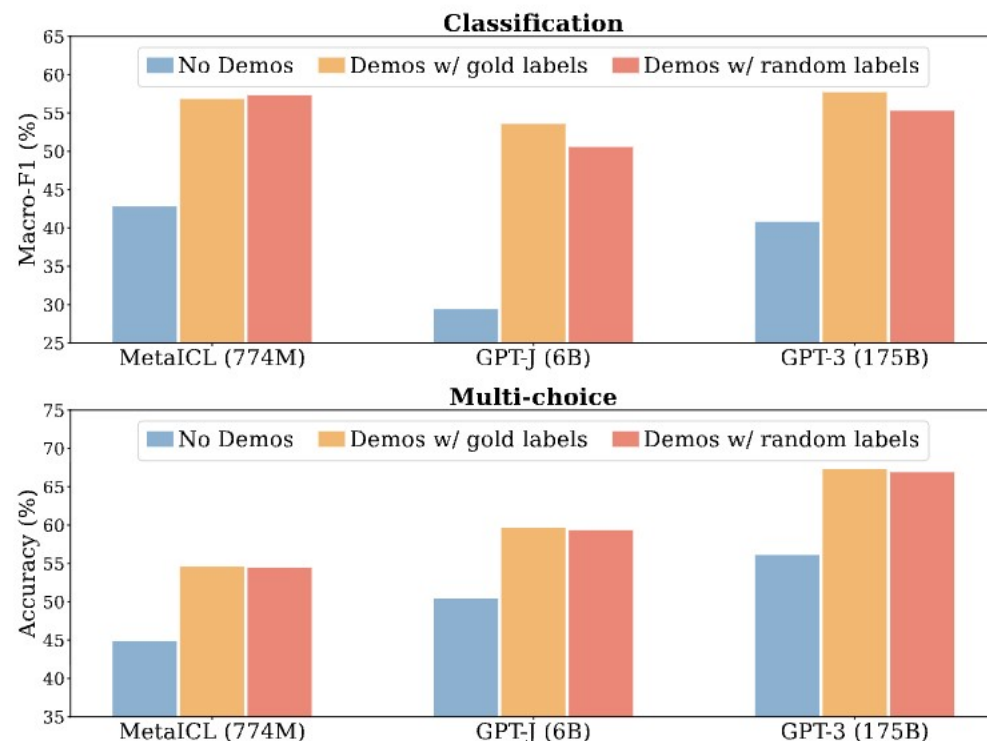


Figure 1: Results in classification (top) and multi-choice tasks (bottom), using three LMs with varying size. Reported on six datasets on which GPT-3 is evaluated; the channel method is used. See Section 4 for the full results. In-context learning performance drops only marginally when labels in the demonstrations are replaced by random labels.

Other observations on In Context Learning

- Formatting correctness means examples should be *exactly* like queries
- Increase number of examples?
 - Diminishing returns in accuracy
 - Increases cost – fills up context
- Larger models use context better
 - <https://arxiv.org/abs/2303.03846>
 - Larger language models do in-context learning differently**

In HW1, the "base" method provided no instructions or examples. It got zero questions right (because it didn't know the alien words) but it was also one of the slowest methods to run, why? ^

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Without formatting instructions it gave long responses	39	80%
✗	The model had to load additional weights for each problem	4	8%
✗	Base prompting disables GPU acceleration	0	0%
✗	The model retried each problem multiple times before giving up	6	12%

What is `glorp(trok(trok(3, 4), flarn(1, 4)))`? (max_token=256)

To solve the expression `\(glorp(trok(trok(3, 4), flarn(1, 4))) \)`, we need to break it down step-by-step. First, let's evaluate the innermost function:

- The first argument of `\( glorp \)` is `\( trok(3, 4) \)`. - According to the problem statement, `\( trok(3, 4) = 7 \)`. Next, we apply this result to the next function: - The second argument of `\( glorp \)` is `\( flarn(1, 4) \)`. - According to the problem statement, `\( flarn(1, 4) = 5 \)`. Now, we combine these results using the outermost function: - The final argument of `\( glorp \)` is `\( \text{glorp}(7, 5) \)`. According to the problem statement, `\( glorp 7, 5) = 20 \)`. Therefore, the value of `\( glorp`
(`trok(trok(3, 4), fl` **Cuts off mid-sentence**

Observations:

- LLM = BS artist without context (makes up info, changes definitions)
- COT is default, format instructions needed to reduce tokens
- COT is verbose
 - if you stop early with "max_tokens" you won't get to the answer

Temperature scaling divides logits by a scalar T before applying softmax. If a model is overconfident (ECE is high because probabilities are too extreme), which direction should you adjust T ?



Answer Frequency Summary

Answer		Respondents	Percentage
✓	$T > 1$	42	86%
✗	$T=1$	1	2%
✗	$T<1$	6	12%
✗	$T=0$	0	0%

We trained with "label smoothing" to improve the calibration. How did we implement this?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	By modifying a hyper-parameter in the cross entropy loss.	35	71%
✗	We modified the labels by dividing by a temperature, T.	9	18%
✗	We randomly flipped some of the labels.	1	2%
✗	We added noise to some of the images.	4	8%

Maybe not a good question?

- **18% answer is kind of reasonable**
- **We didn't implement anything from scratch so no reason to remember this**

```
nn.CrossEntropyLoss(label_smoothing=0.1)
```

The Missing Link: ML on Graphs

Slides from former PhD student:

Sami Abu-El-Haija

(Google Research -> vibe coding startup founder!?)

Why should you pay attention today?

Most practical data is **relational**. Examples don't exist in isolation.

- Users write (**NLP**) comments, upload (**Image**) photos, download apps, etc.
- While different models handle different modalities ...
 - How can we combine information across these relationships?
 - ML models on Graphs, e.g., Graph Neural Networks.

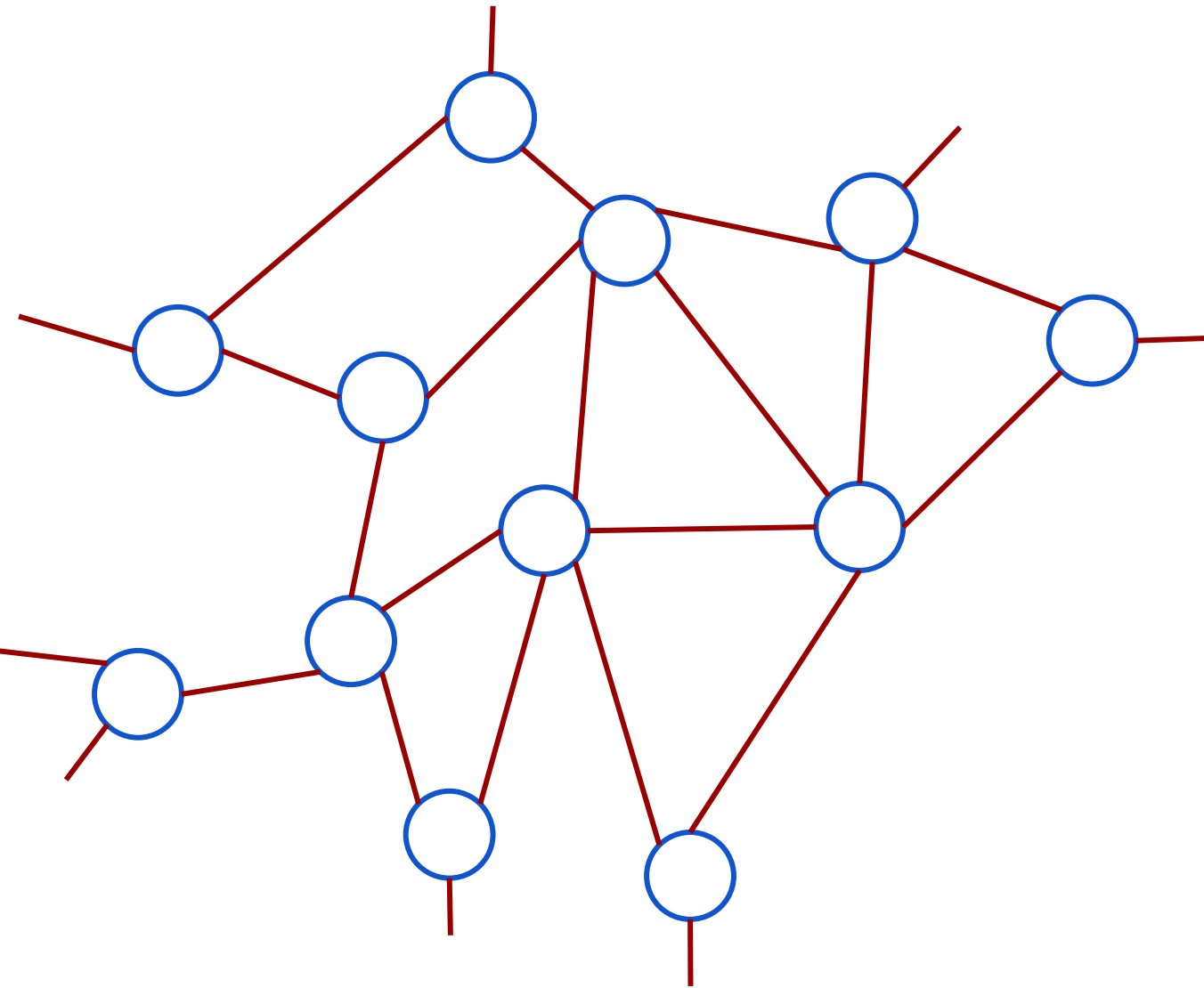
Agenda

- What is a Graph
- ML Tasks on Graphs
- Model Families for ML on Graphs
- Graph Neural Networks
 - Background
 - Practical Pitfalls (modeling & scale)
 - Addressing Pitfalls

Presentation prefers breadth over depth

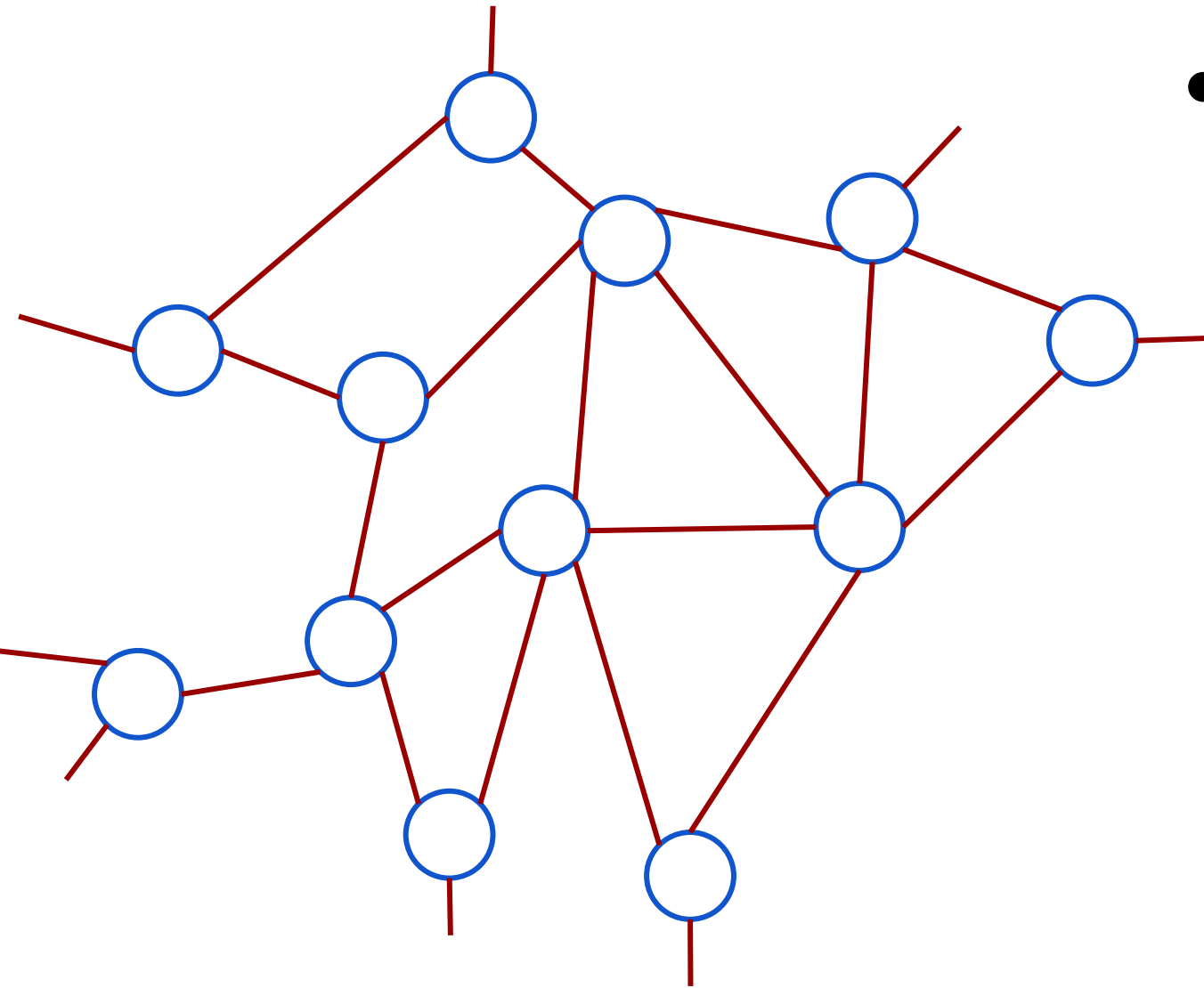
What is a Graph

What is a graph?

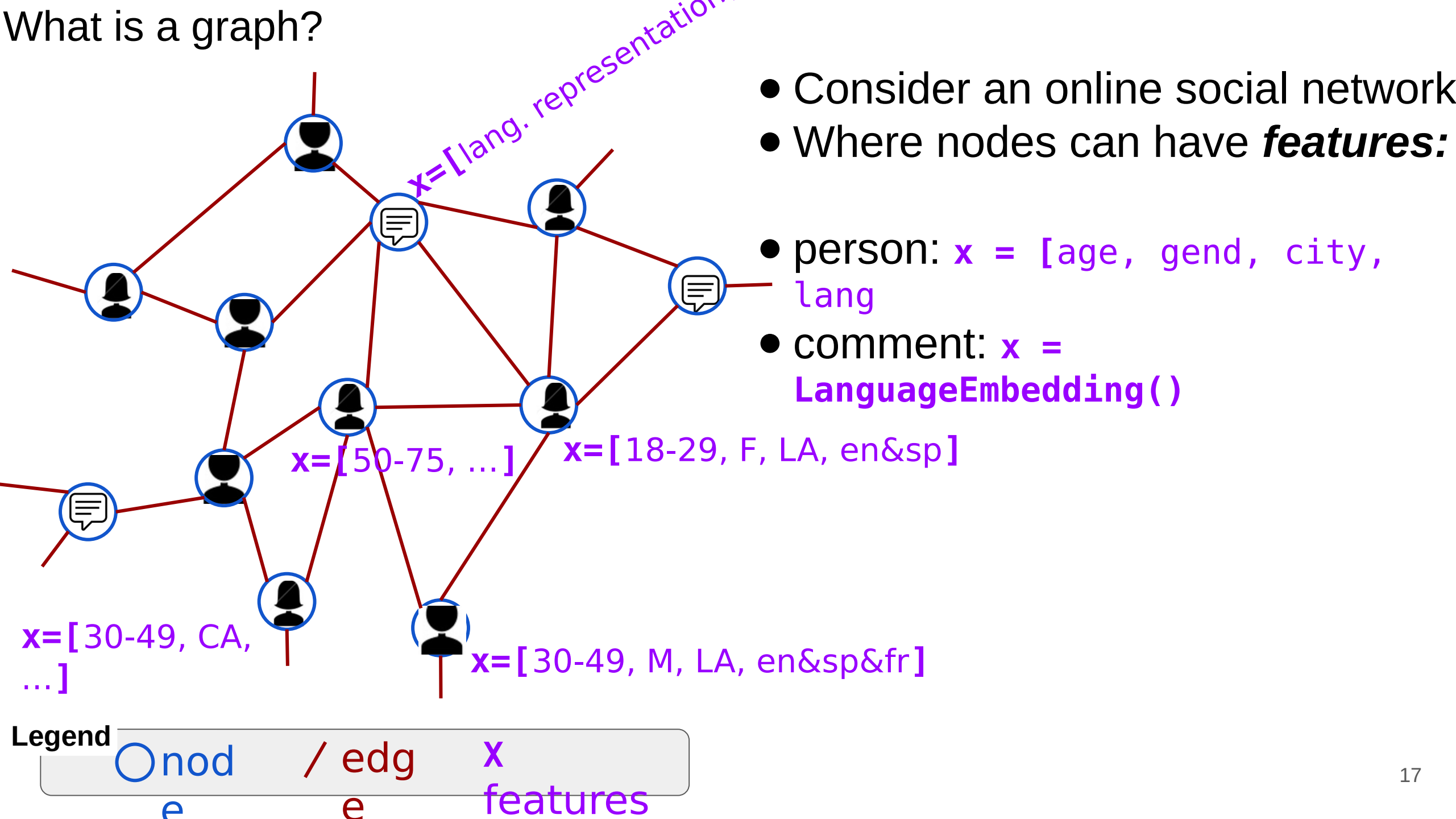


What is a graph?

- Consider an online social network



What is a graph?

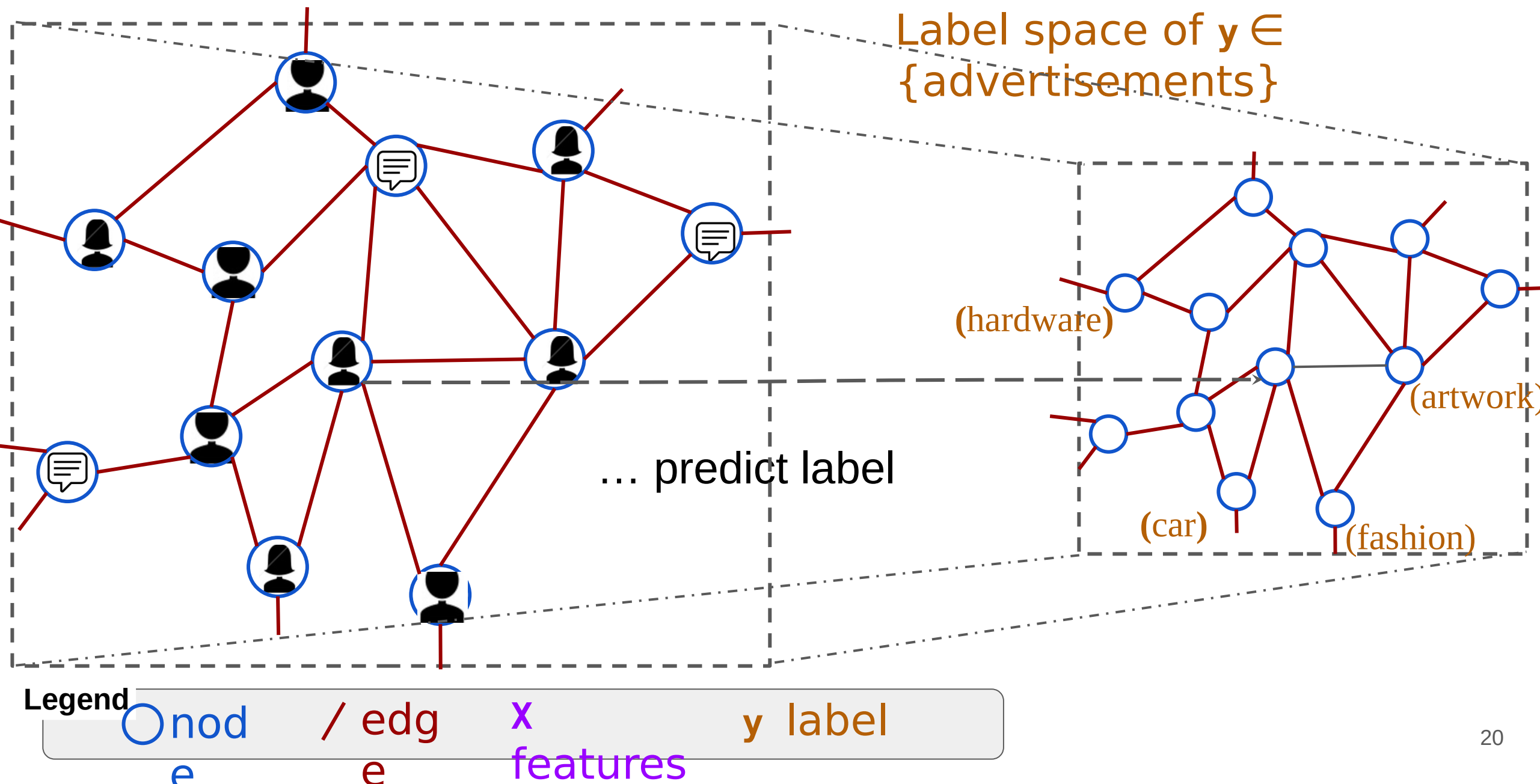


ML Tasks on Graphs

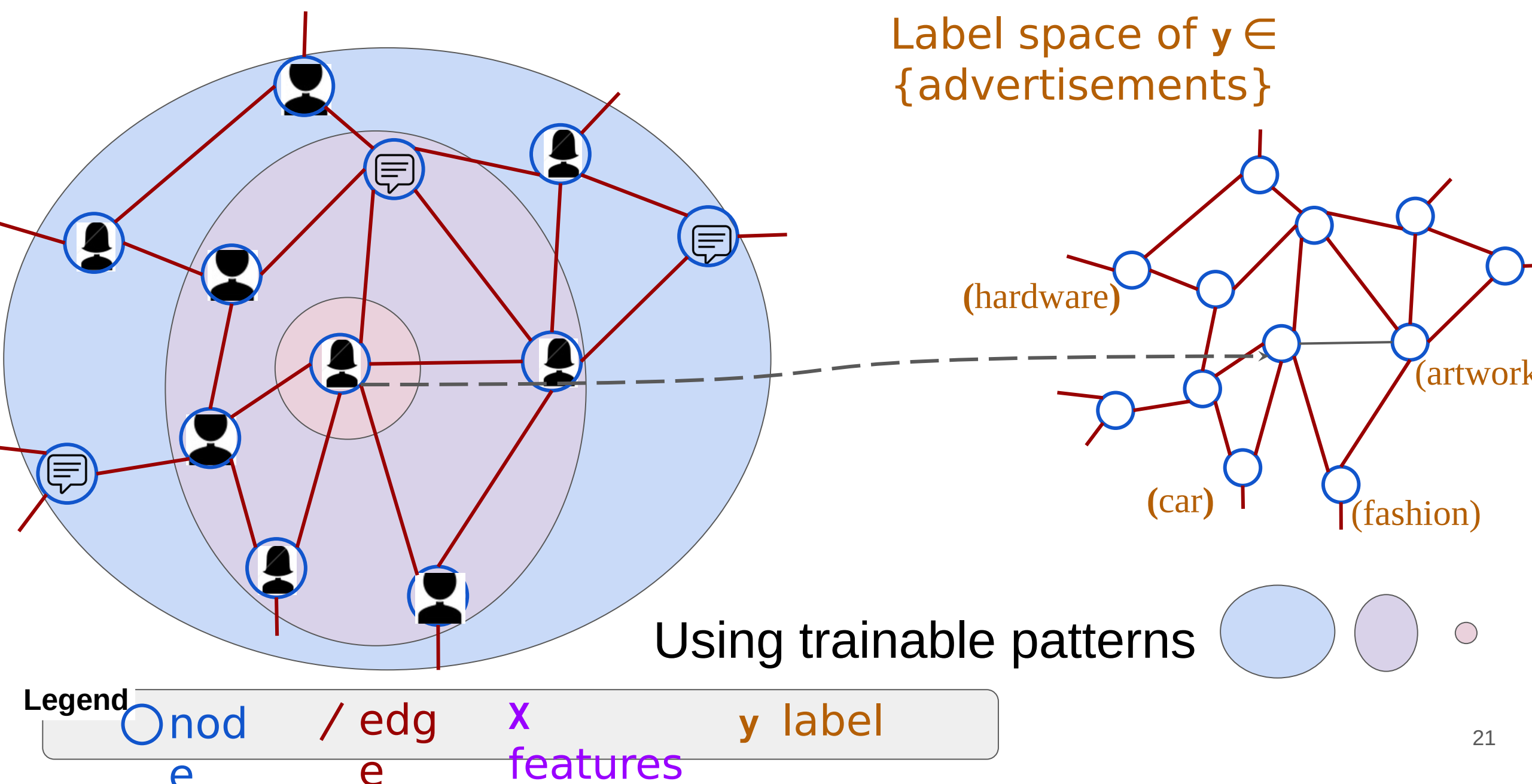
Focus: Node Classification & Link Prediction

Machine Learning on Graphs can do **Node Classification**

Machine Learning on Graphs can do Node Classification

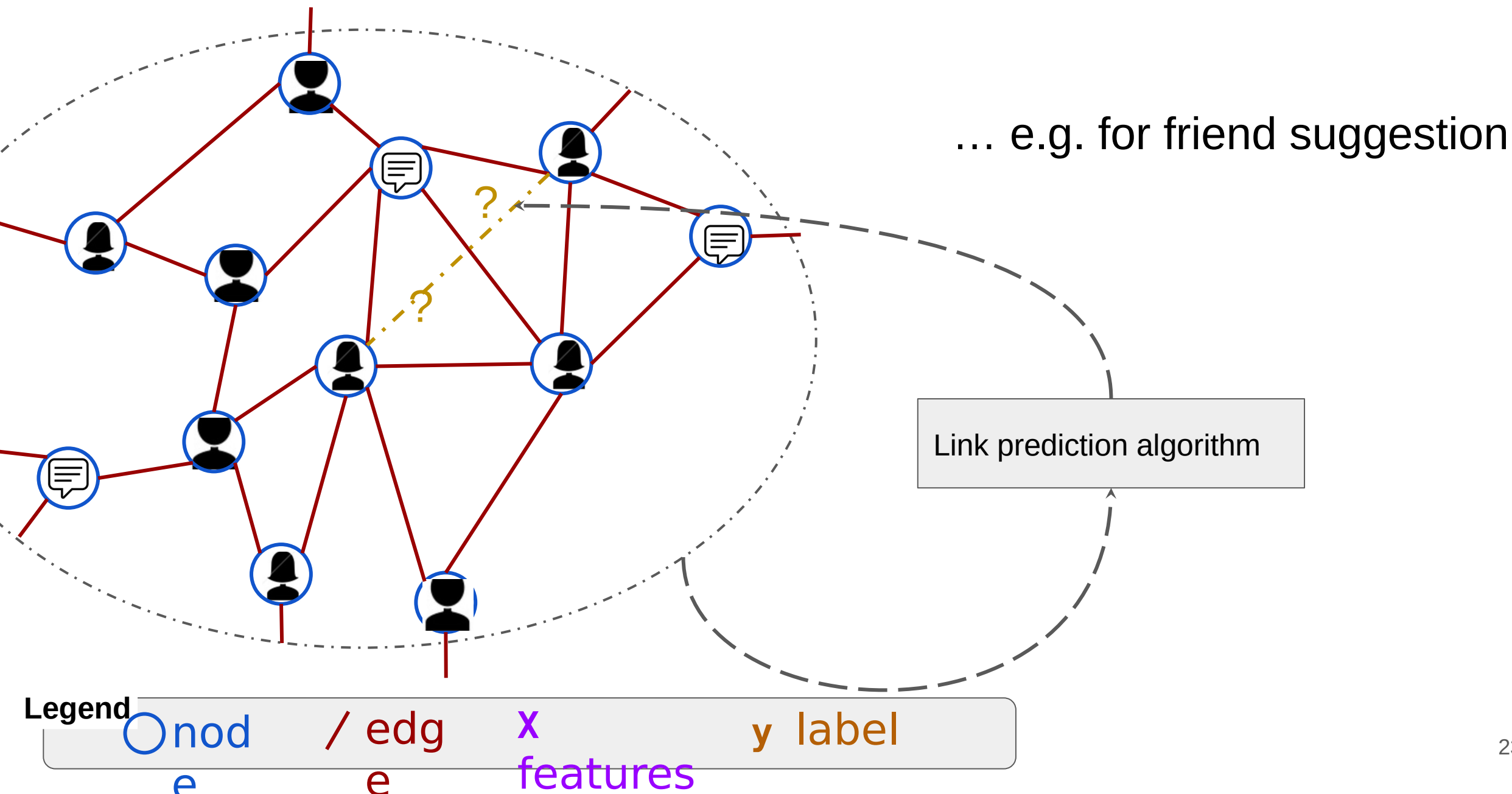


Machine Learning on Graphs can do Node Classification



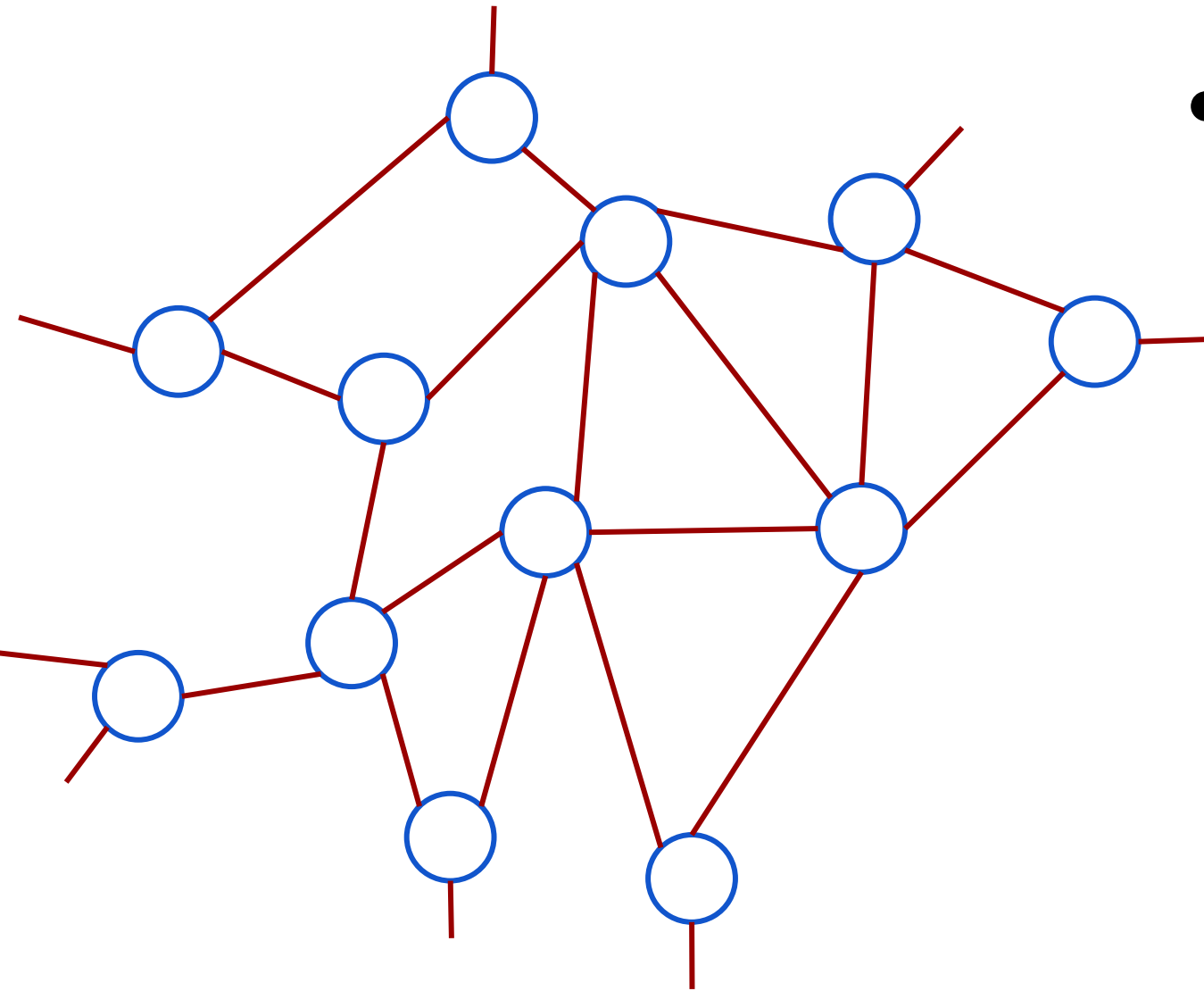
Machine Learning on Graphs can do **Link Prediction**

Machine Learning on Graphs can do Link Prediction



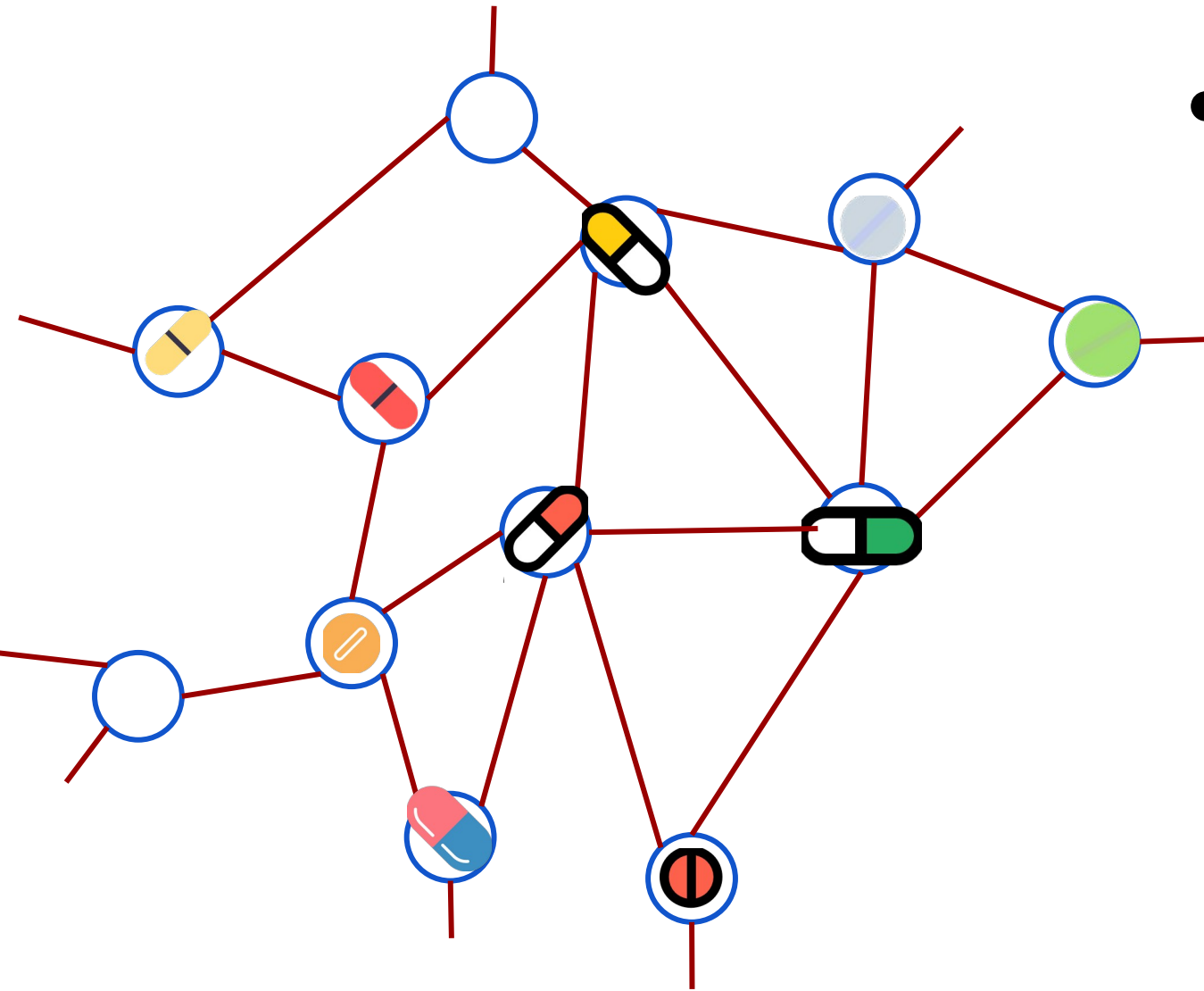
Link Prediction on drug-drug interaction networks

- Consider drug-drug interactions



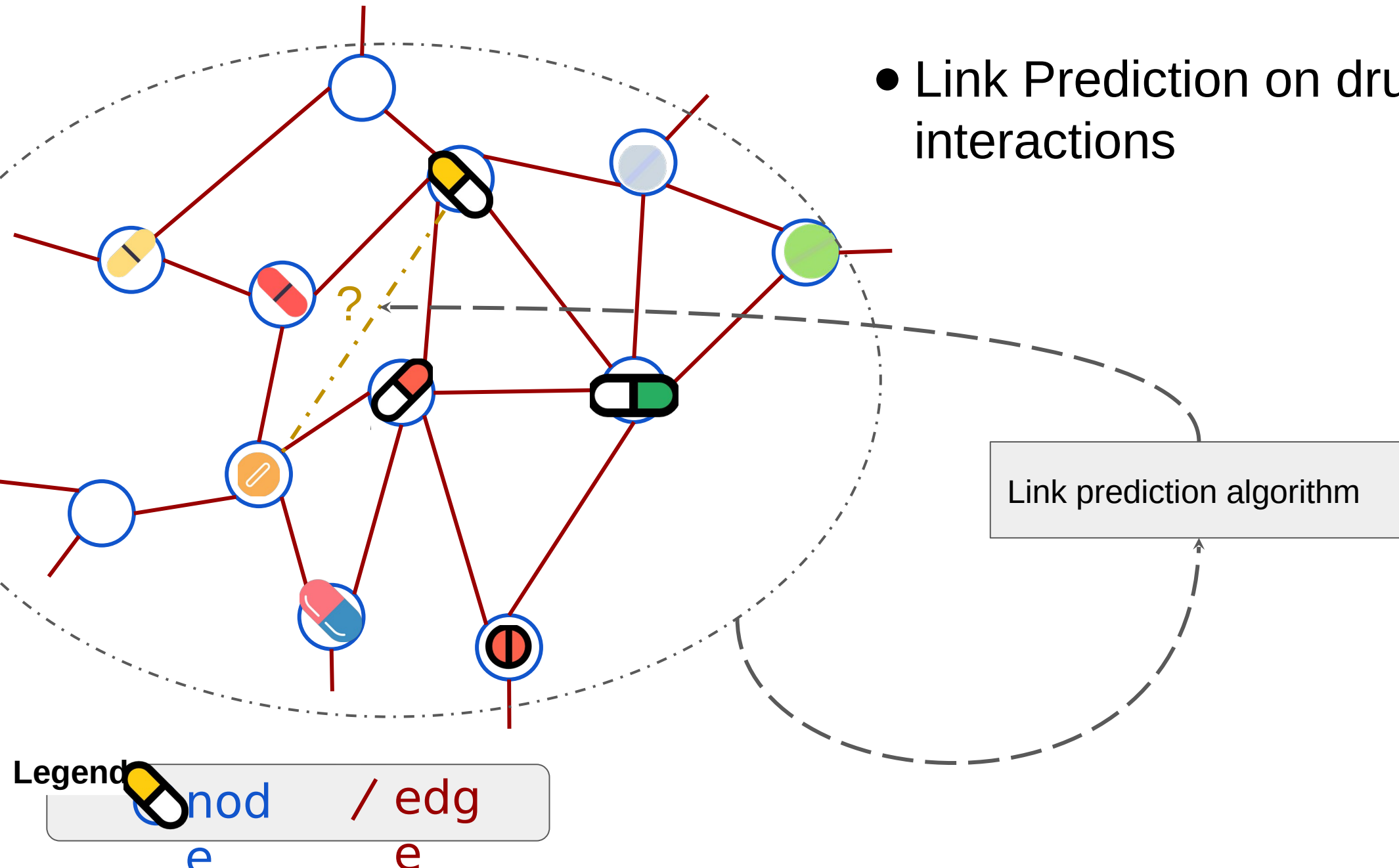
Link Prediction on drug-drug interaction networks

- Consider drug-drug interactions

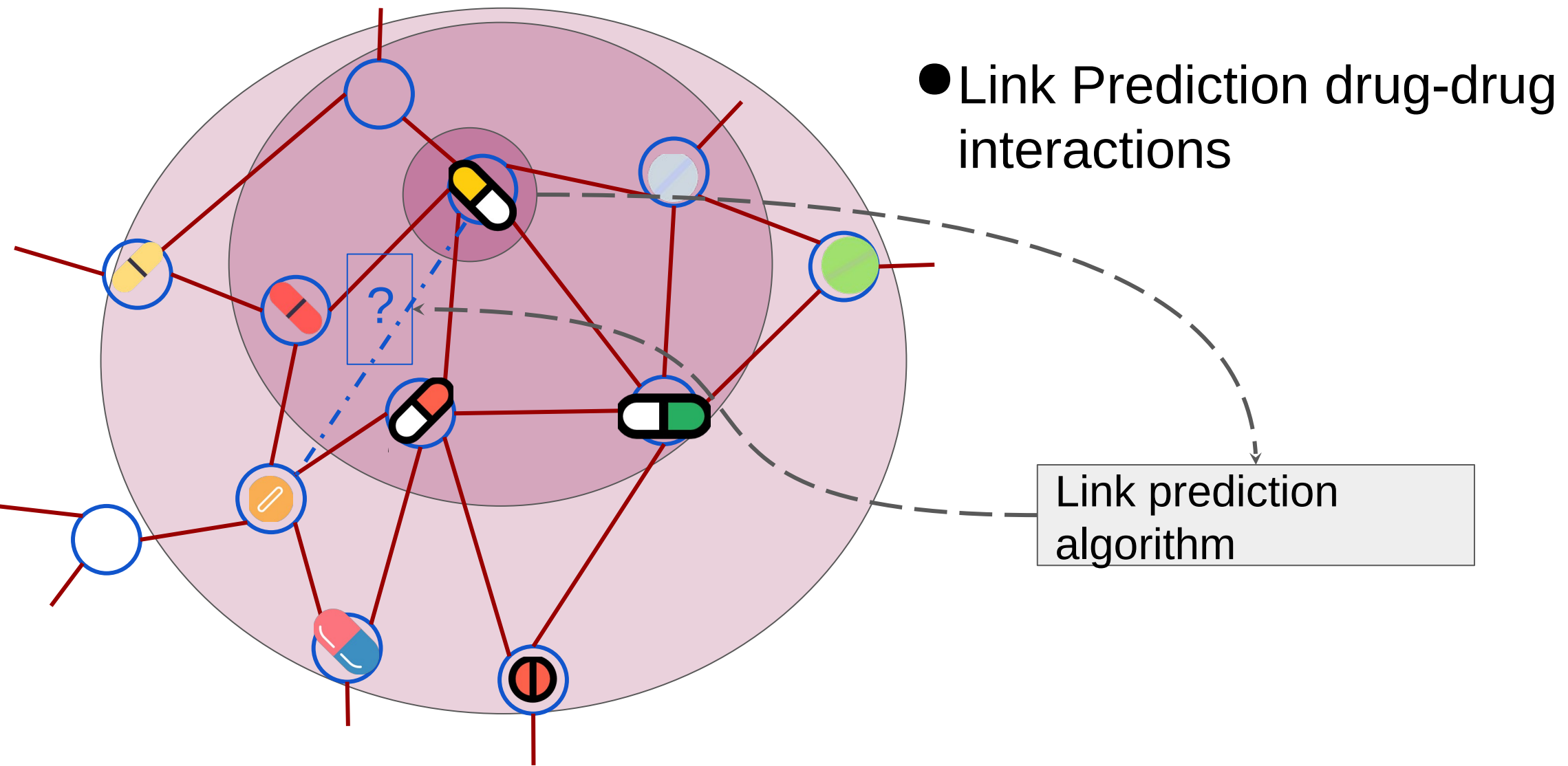


Link Prediction on drug-drug interaction networks

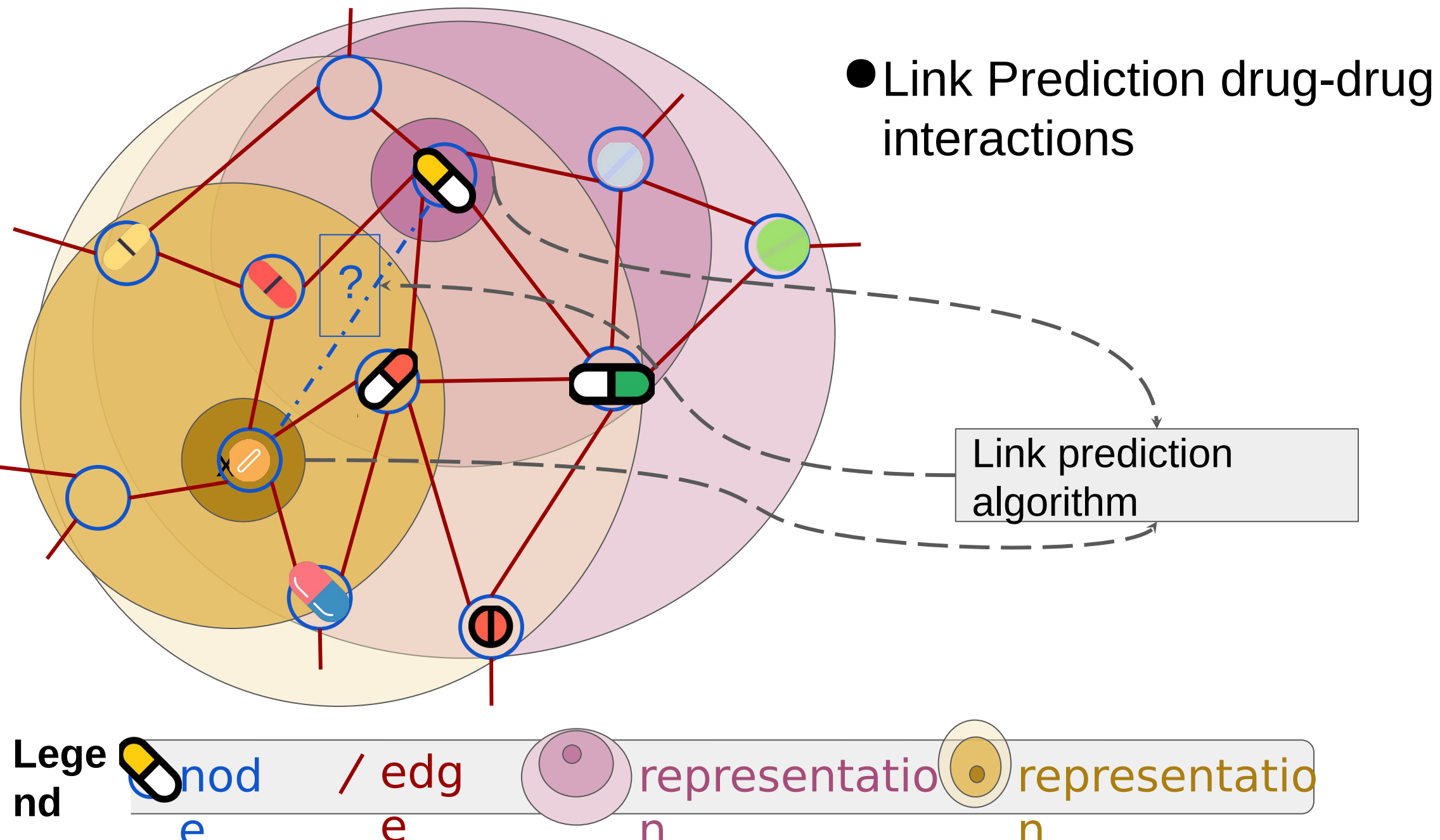
- Link Prediction on drug-drug interactions



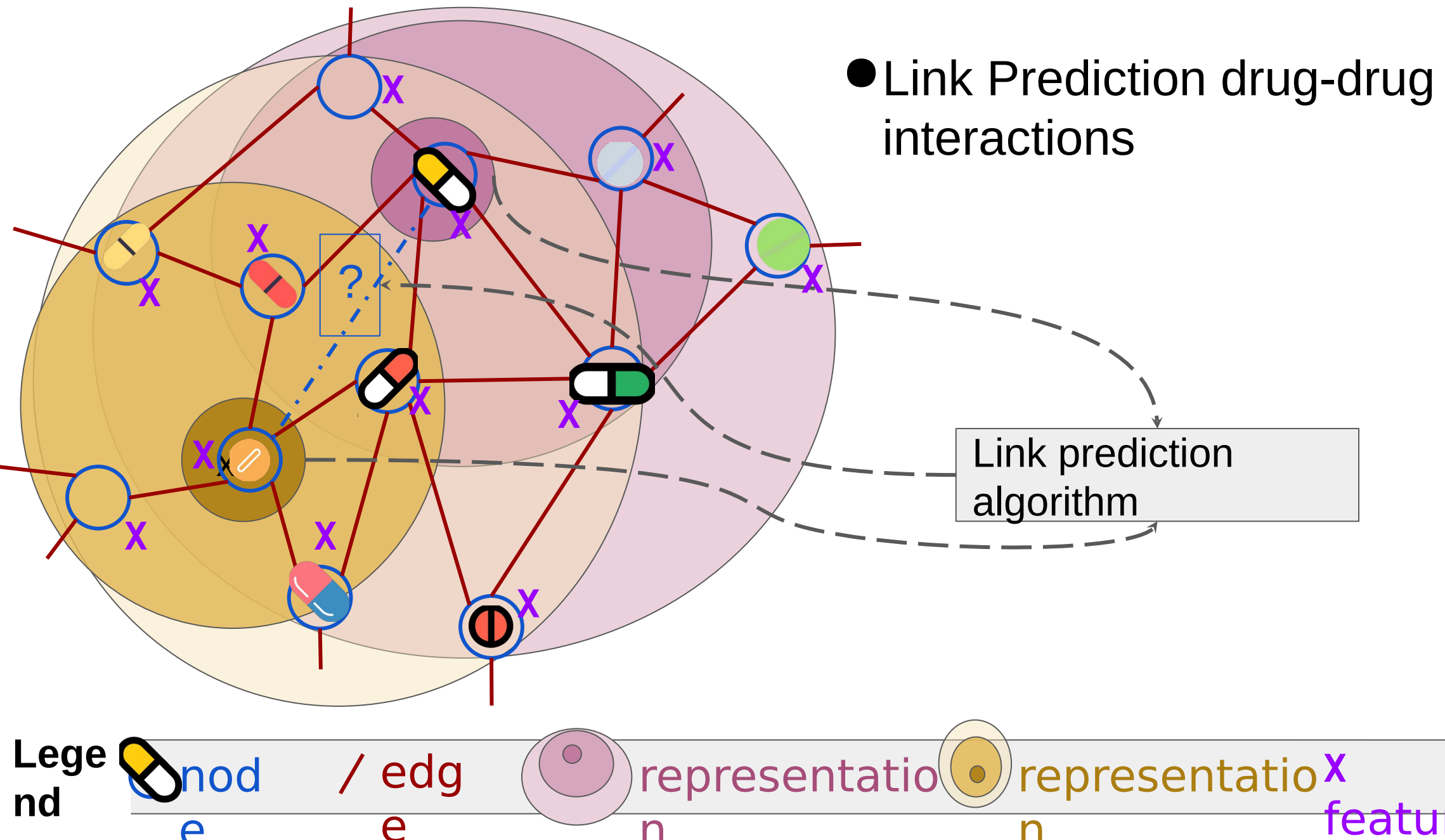
Link Prediction on drug-drug interaction networks



Link Prediction on drug-drug interaction networks



Link Prediction on drug-drug interaction networks



Why ML on Graphs (Motivation)

Graphs are general data structures. Algorithms are portable across domains.

- Task of Link Prediction and Node Classification apply to various domains.

Dataset	Node	Edge	Utility of Link Prediction	Utility of Node Classification
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Why ML on Graphs (Motivation)

Graphs are general data structures. Algorithms are portable across domains.

- Task of Link Prediction and Node Classification apply to various domains.

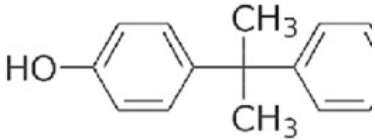
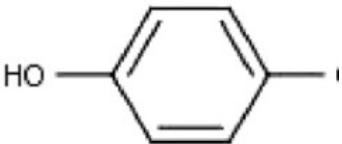
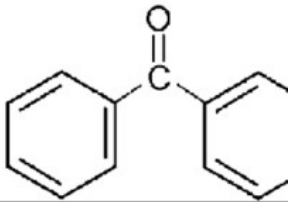
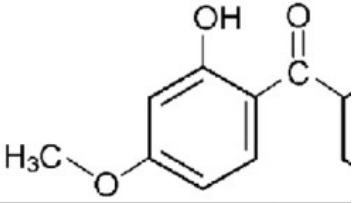
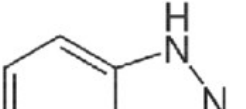
Dataset	Node	Edge	Utility of Link Prediction	Utility of Node Classification
Social Network	Person	Friendship	Friend Suggest	Ad preference
Netflix	User or Video	“Like”	Video Recommendation	What is this video about?
Drug-Drug interactions	Drug	interaction	Do 2 drugs interact?	Predict Drug bio{physical, chem} (e.g., toxicity, boiling temperature, antigen)
Citation Network	Article	Citation	Are articles related? (should one have cited the other)	Article Categorization

ML Tasks on Graphs

Beyond Node Classification & Link Prediction

ML on Graphs can do Graph-level Prediction

- Biochemistry

Compound	CAS number	Formula	Chemical Structure
Bisphenol A (BPA)	80-05-7	$C_{15}H_{16}O_2$	
Nonylphenol (NP)	104-40-5	$C_{15}H_{24}O$	
Benzophenone (BP)	119-61-9	$C_{13}H_{10}O$	
Benzophenone-3 (BP-3)	131-57-7	$C_{14}H_{12}O_3$	
Benzotriazole (BT)	95-14-7	$C_6H_5N_3$	

- Computer Security
 - Computer programs are “Graphs”.
 - Task:** Does a given program contains malware?

Task: output graph-level prediction, e.g.,
boiling temperature, binds to ProteinA?,...

Summary of Feed-forward Tasks -- $P(Y \mid \text{Graph})$

Input: Graph (nodes, edges, features)

Output:

- Graph-level (e.g., Graph Regression)
- Node-level (e.g., Node Classification)
- Node-pair level (e.g., Link Prediction)

inductive: test on new graphs

Can be inductive: test on a new graph

Can be transductive: test on training graph.

Can be on Homogeneous or Heterogeneous Graphs

Generative Tasks -- $P(\text{Graph} \mid Y)$

Input: Attribute

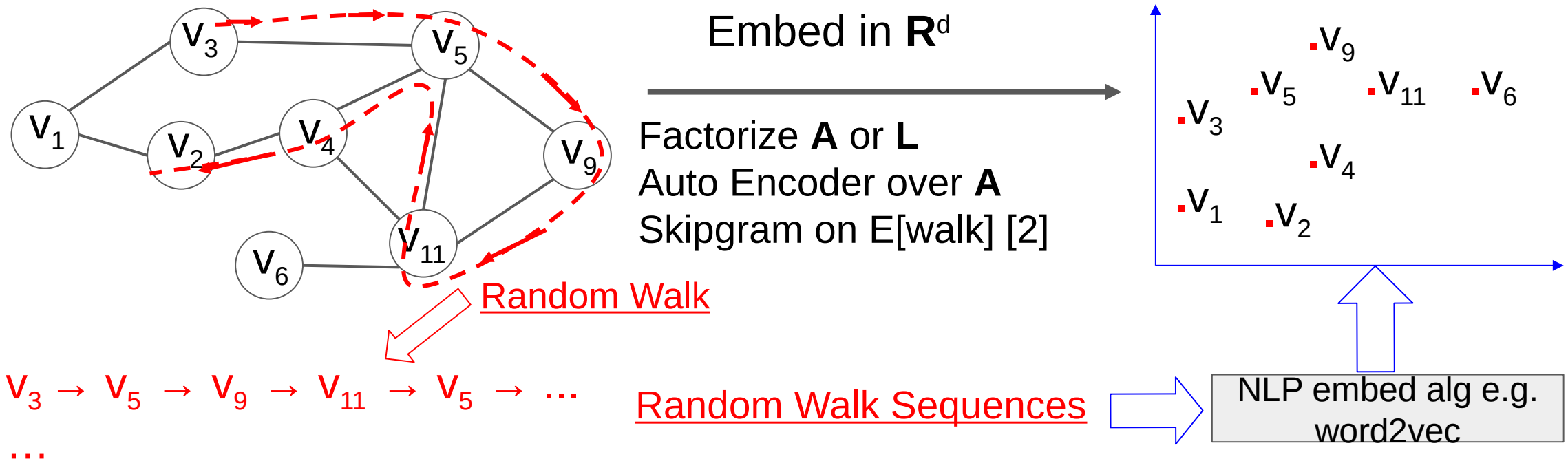
Output: Graph

E.g., drug synthesis

Will not cover today

Model Family:
Skip-gram based
(most are transductive)

Skip-gram Based: embed each node



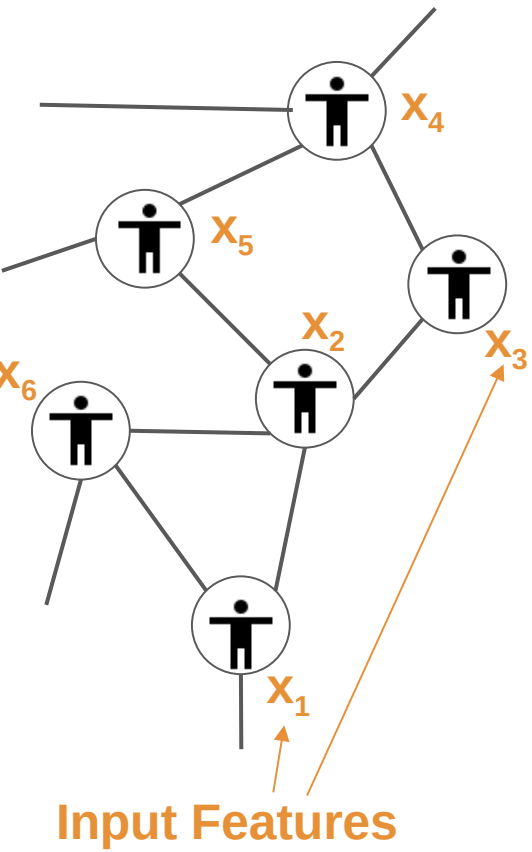
Perozzi et al, KDD 2016

$\mathbf{X}$ node features (n, d) ; $\mathbf{A}$ adjacency matrix (n, n)

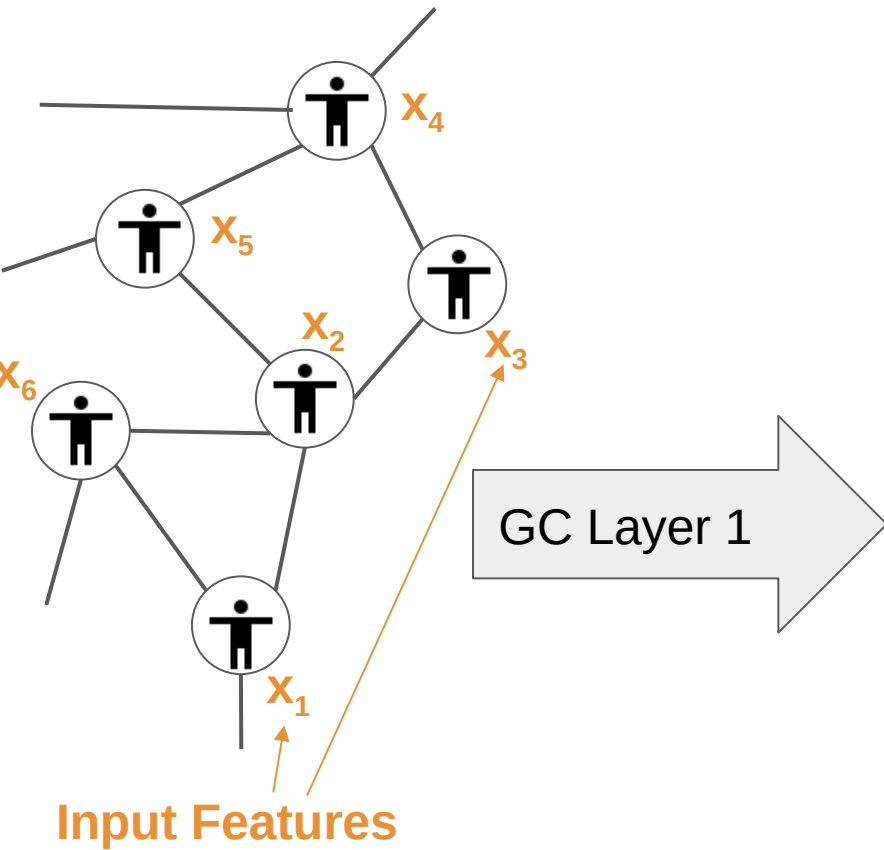
$\mathbf{X}_i$ feature for node i $\mathbf{A}_{i,j} = 1$ iff $i-j$ are connected

Model Family:
Message-passing based
(most are inductive)

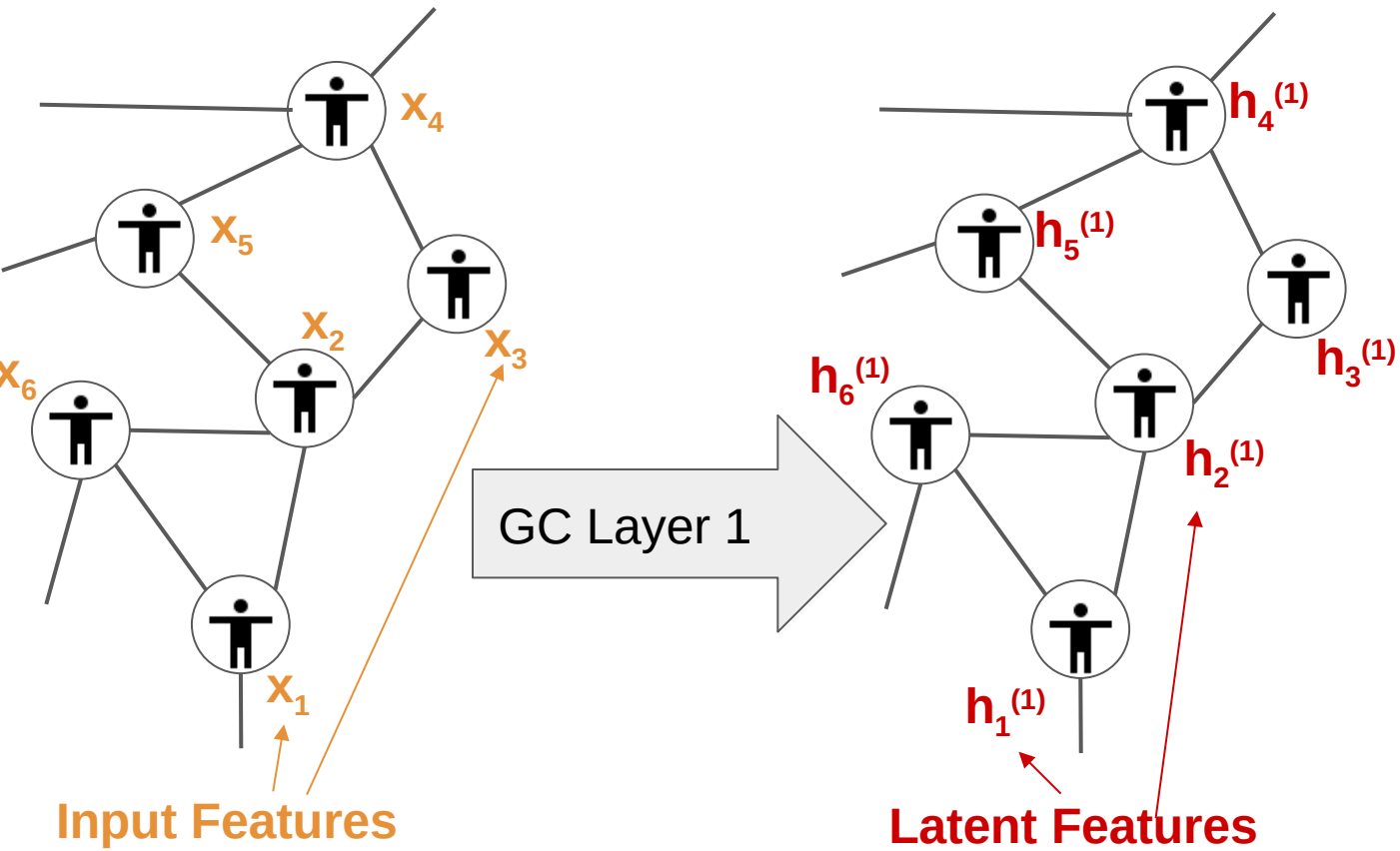
Graph Convolutional Network (GCN)



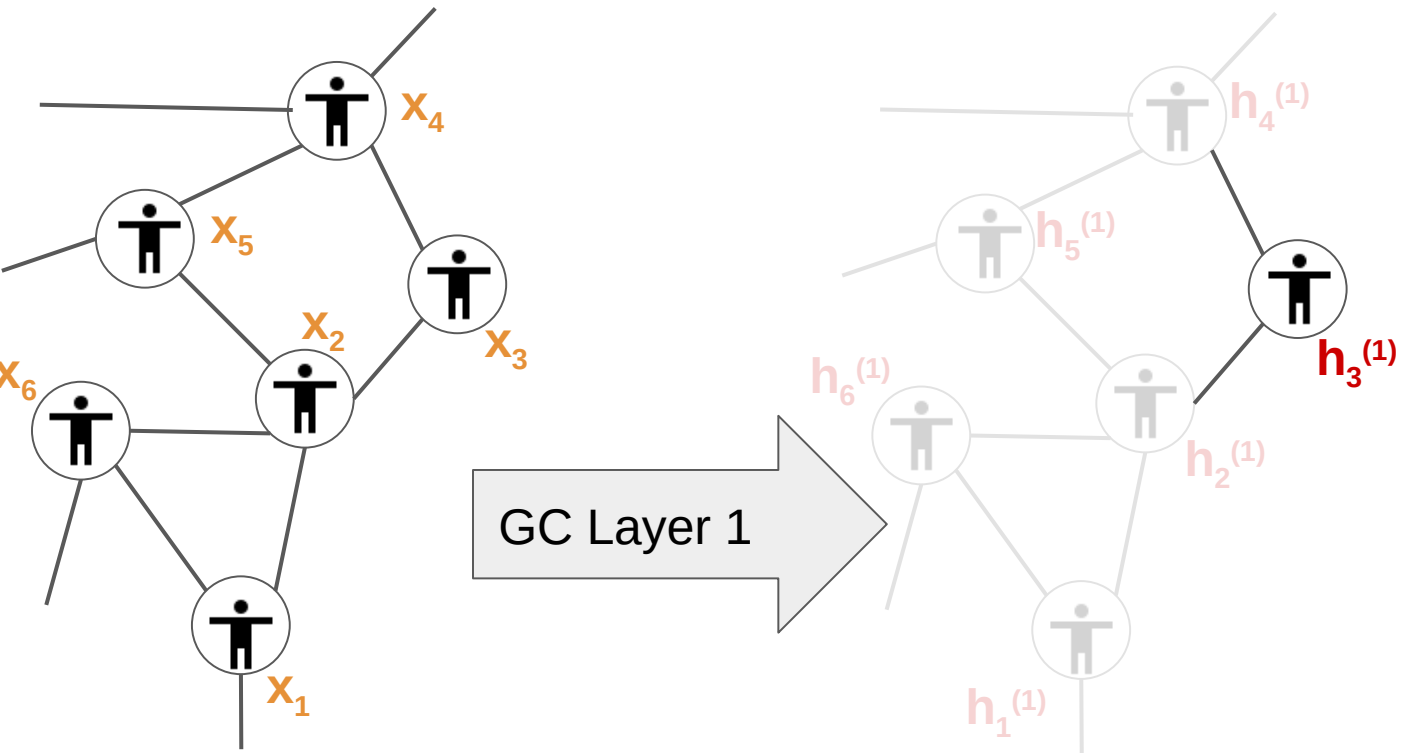
Graph Convolutional Network (GCN)



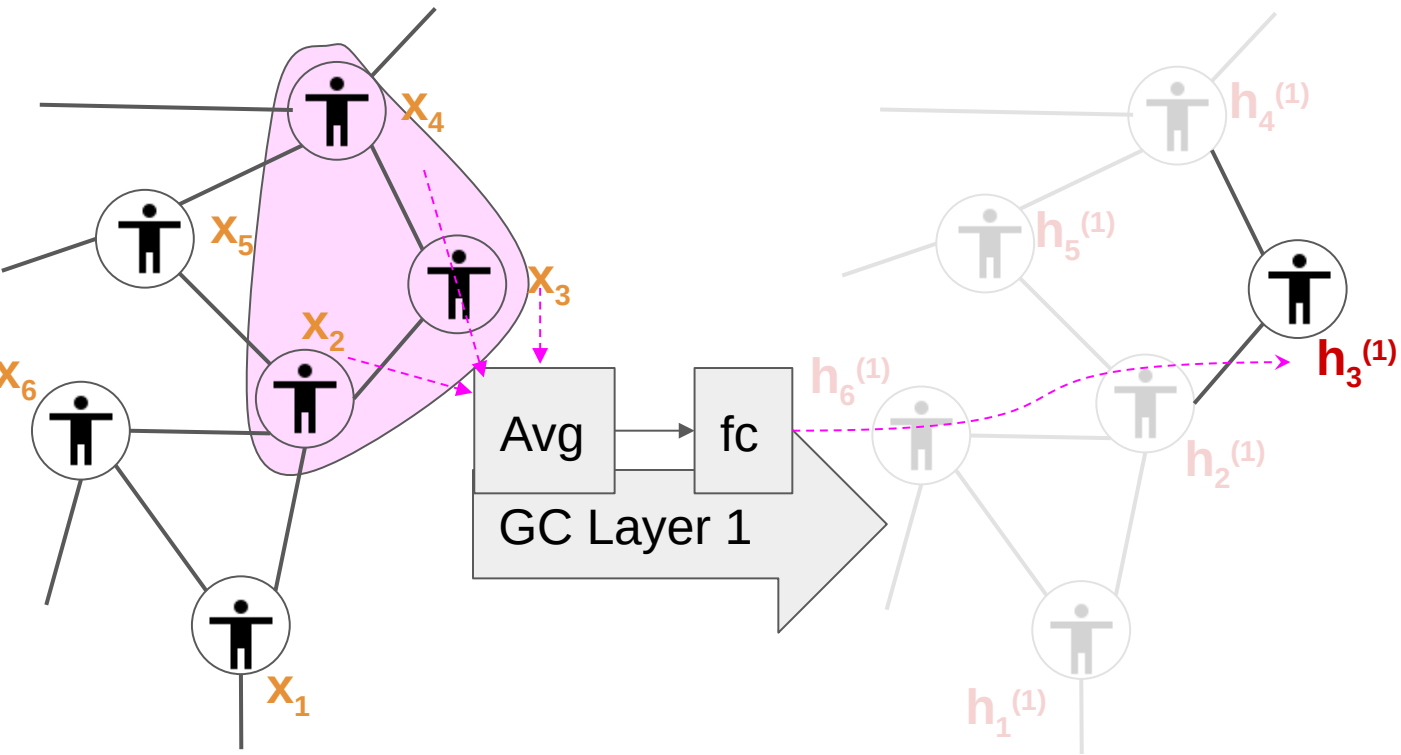
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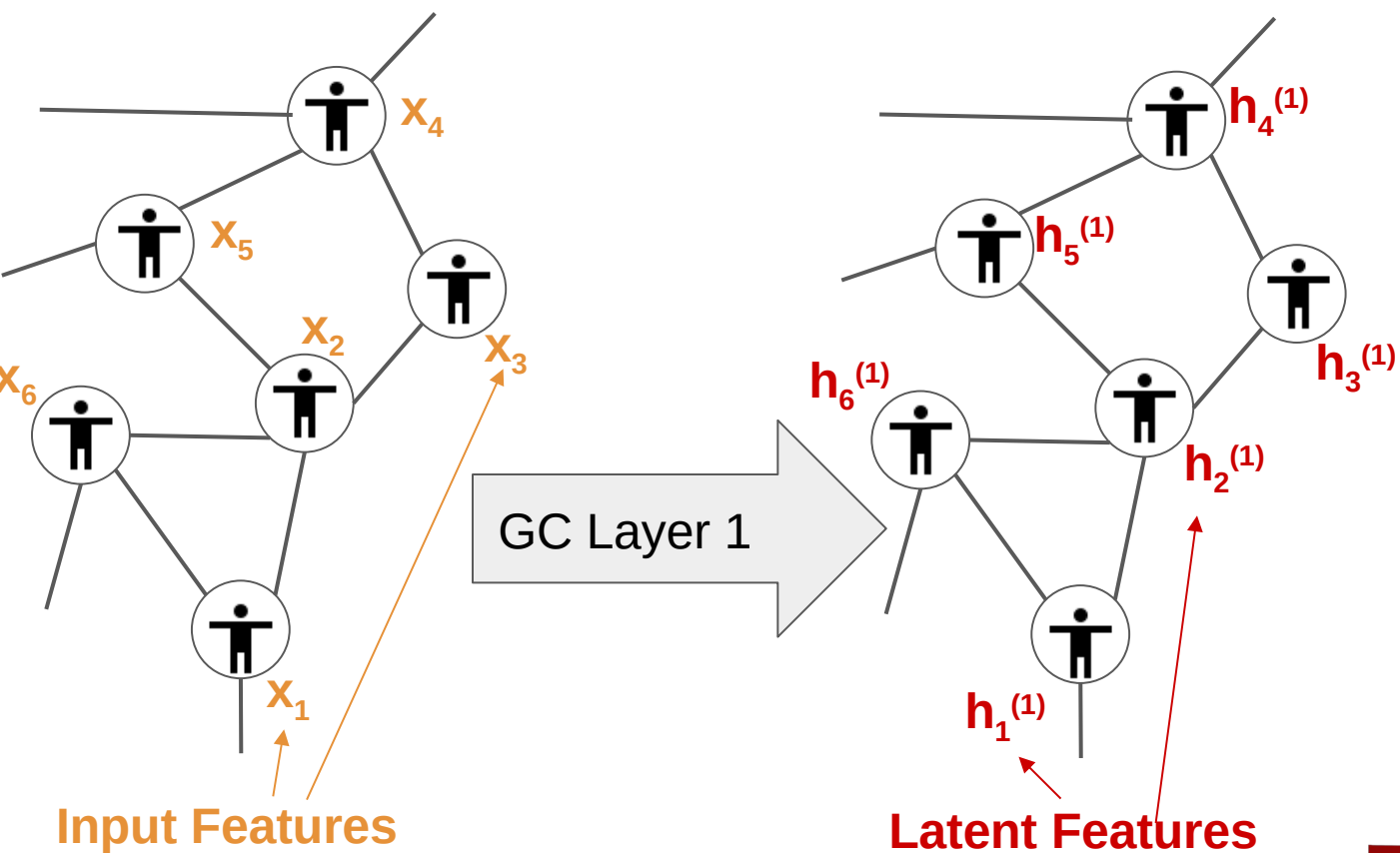
Graph Convolutional Network (GCN)



Graph Convolutional Network (GCN)



Graph Convolutional Network (GCN)



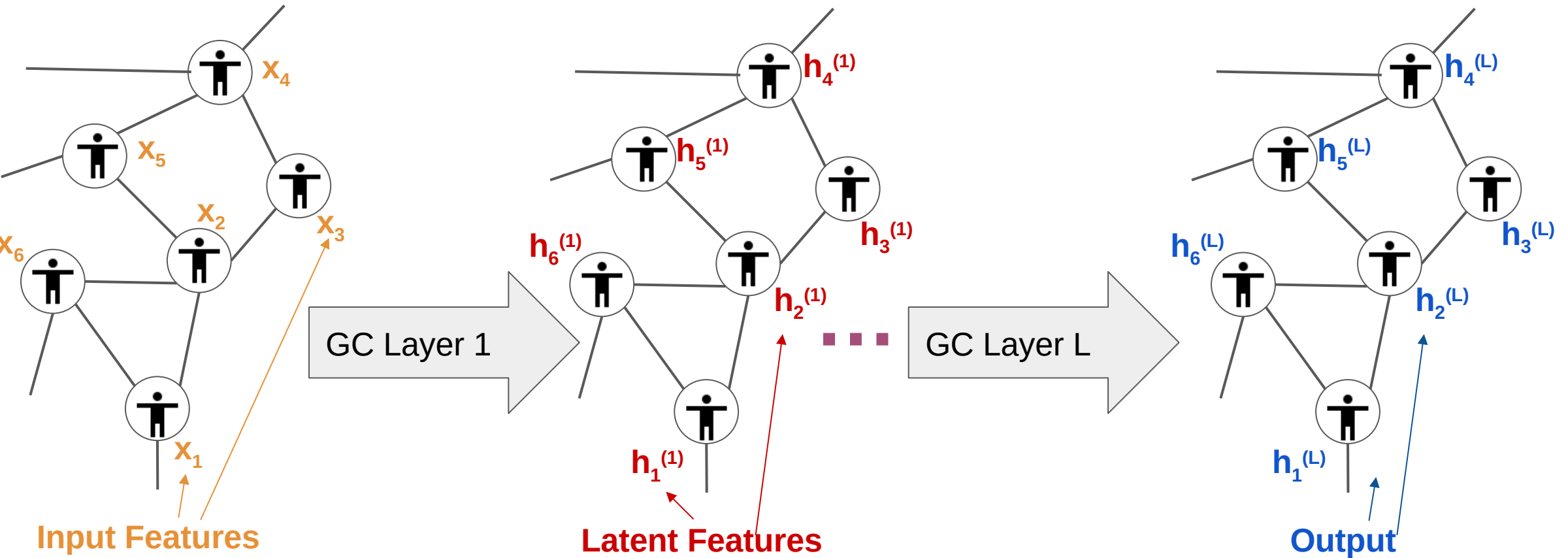
Note: Averages every node features with its 2-step neighborhood

$$\widehat{A}^2 X$$

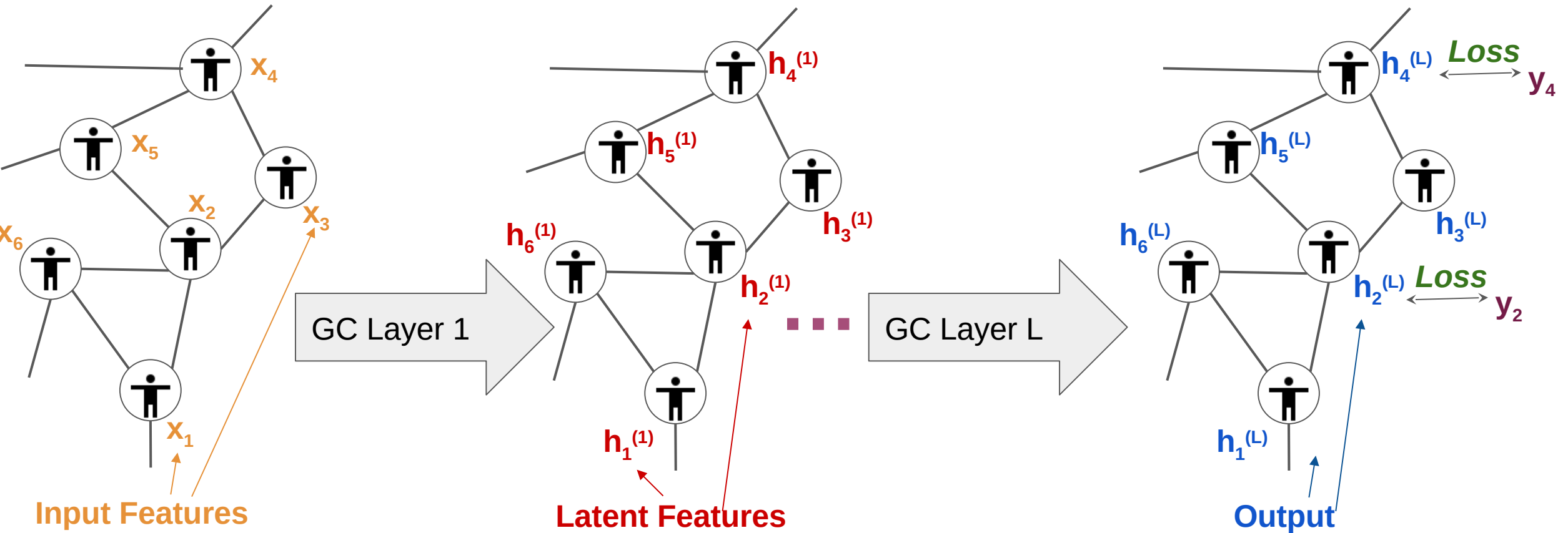
Averages every node's features with its neighbors

$$H^{(1)} = \text{fc}(\widehat{A} X)$$

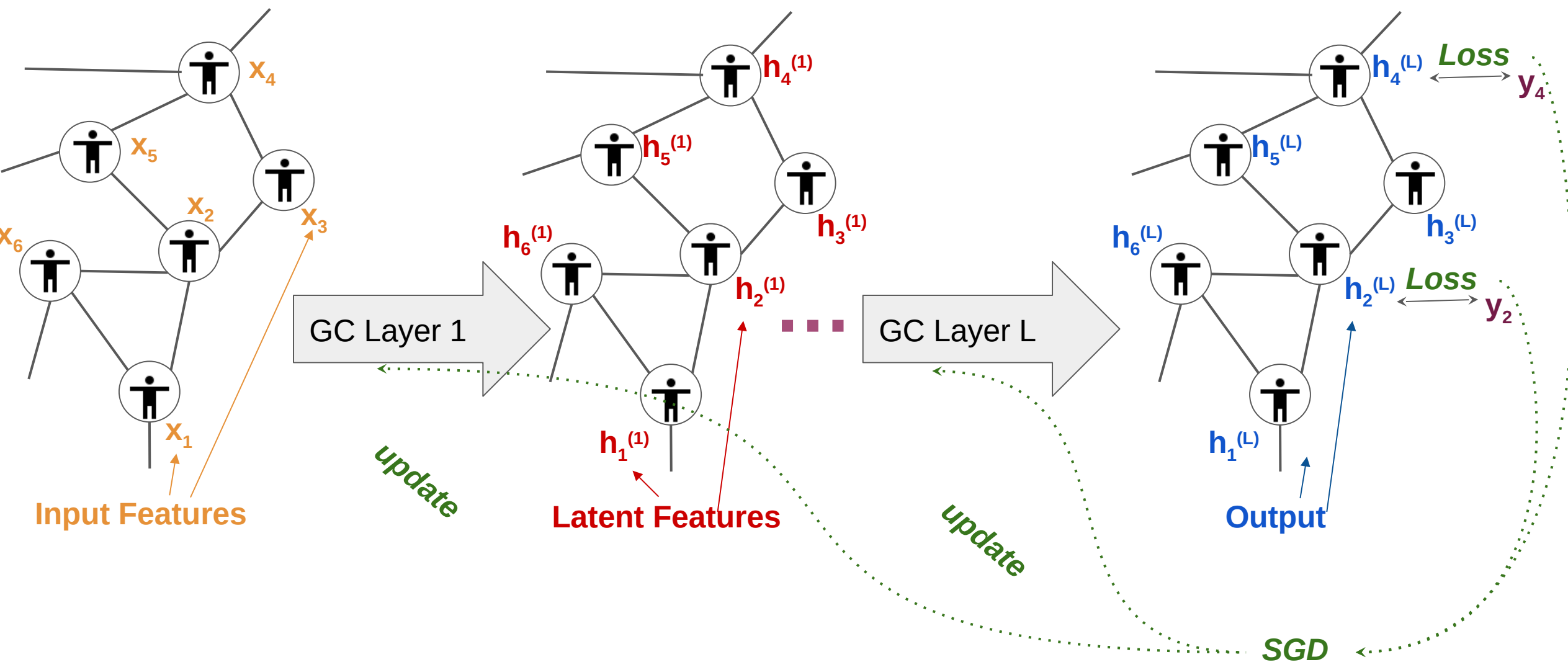
Graph Convolutional Network (GCN)



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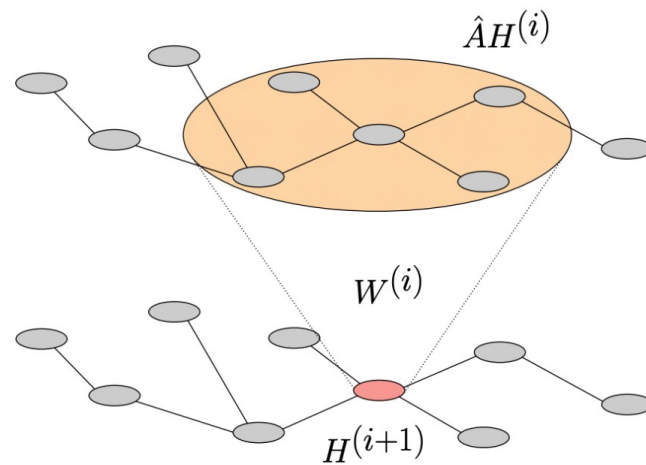
Pitfalls

- Modeling Capacity
 - Adding more GNN layers “over-smoothes” the representation.
 - With many layers, each node will have the exact same representation **(eventually averages whole graph)**
- Scalability
 - Early GNNs are formed on the graph as a whole **(No iid samples – no way to get batches?)**

Addressing Pitfall #1: Modelling Capacity

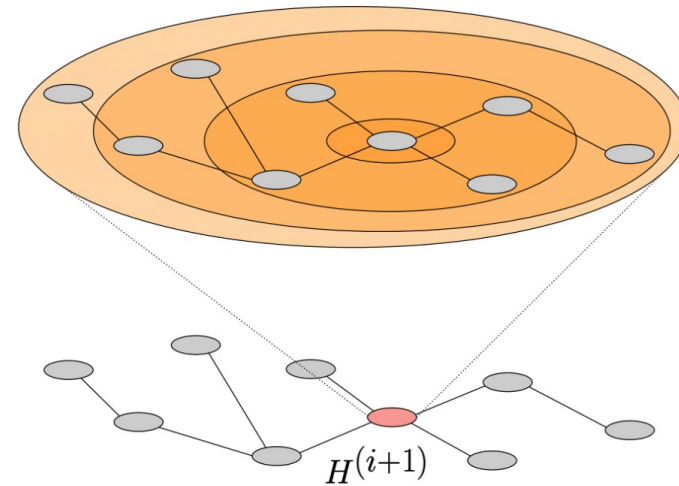
MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

Sami Abu-El-Haija¹ Bryan Perozzi² Amol Kapoor² Nazanin Alipourfard¹ Kristina Lerman¹
Hrayr Harutyunyan¹ Greg Ver Steeg¹ Aram Galstyan¹



$$H^{(i+1)} = \sigma(\hat{A}H^{(i)}W^{(i)})$$

(a) Traditional graph convolution.

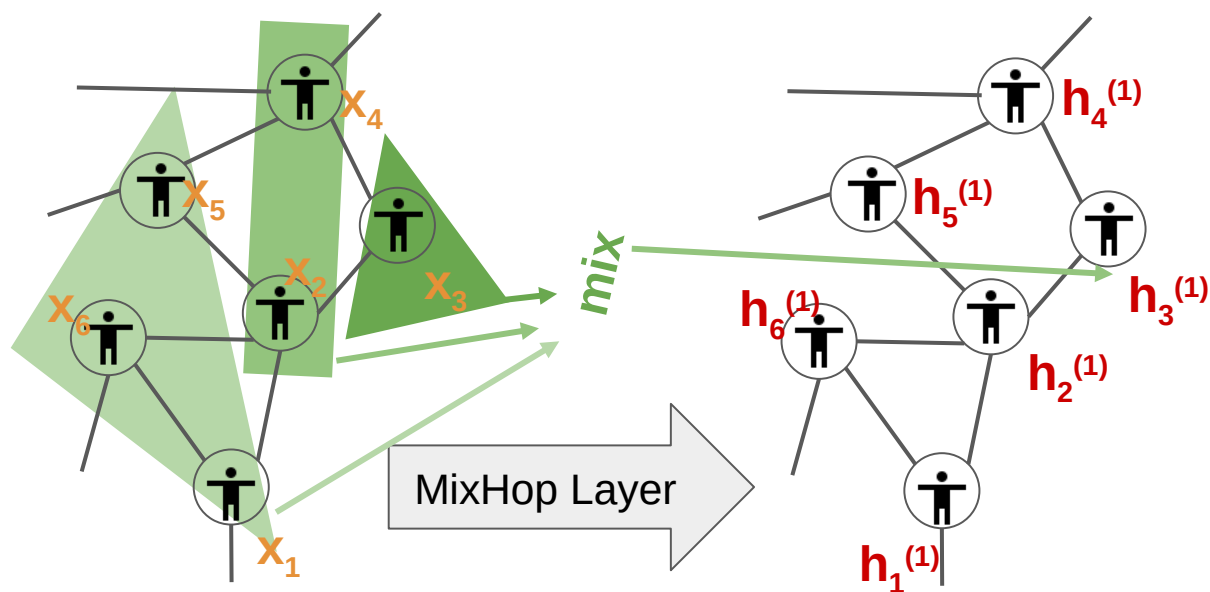


$$H^{(i+1)} = \sigma\left(\hat{A}^0 H^{(i)} W_0^{(i)} \parallel \hat{A}^1 H^{(i)} W_1^{(i)} \parallel \dots\right)$$

(b) Our mixed feature model.

MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

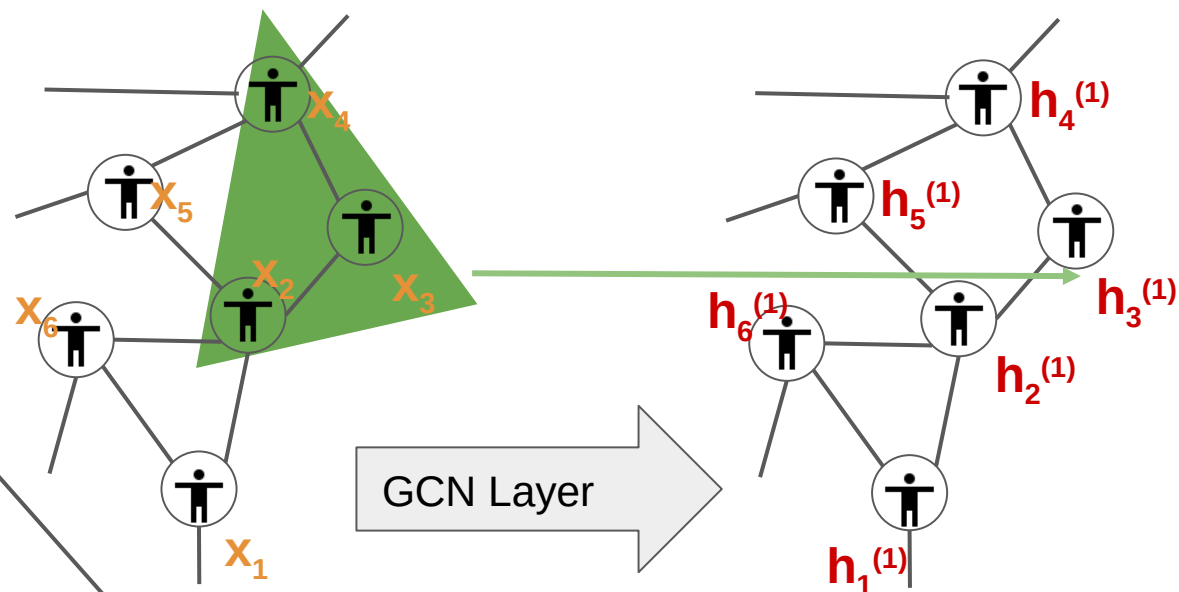
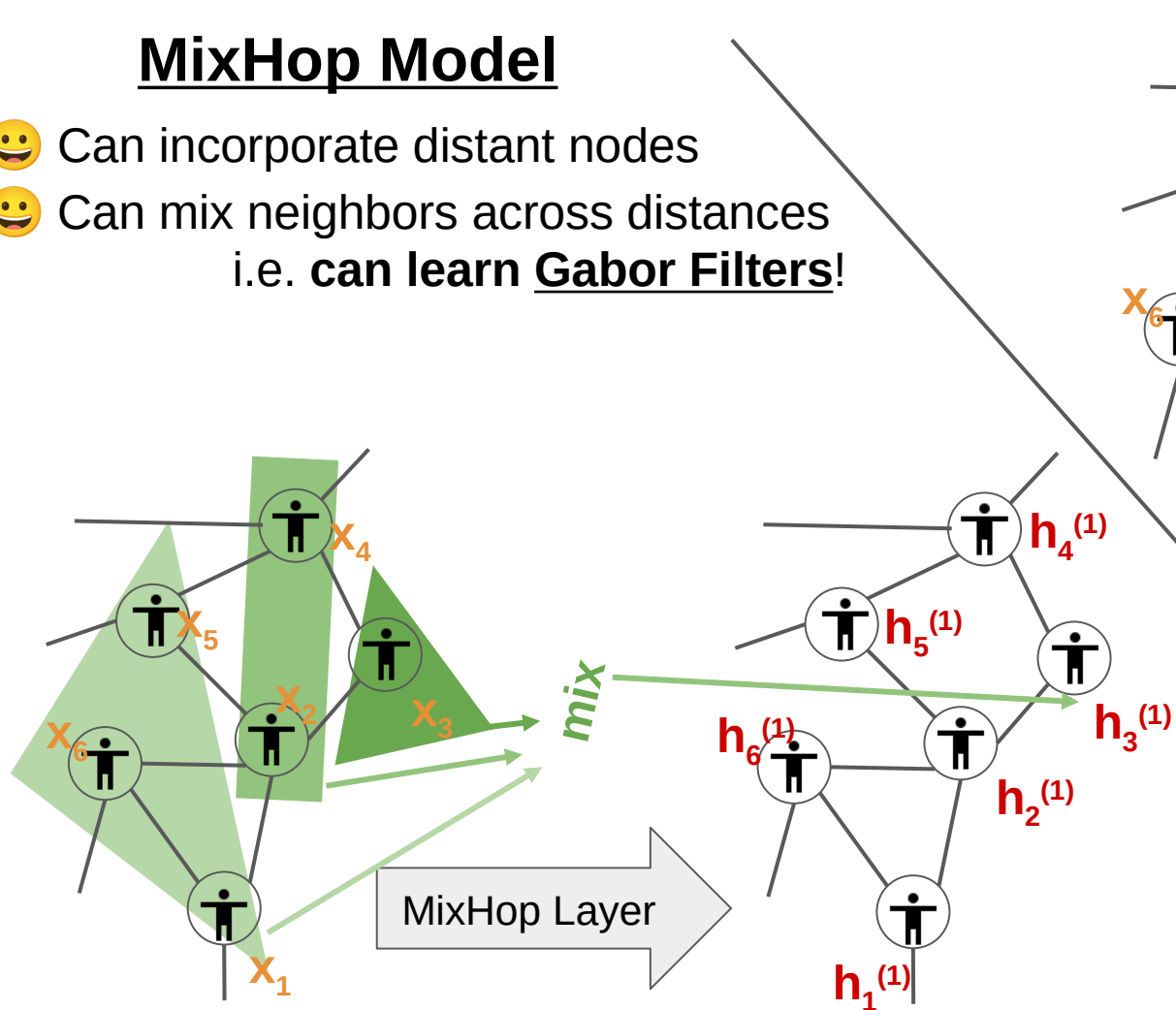
MixHop Model



MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

MixHop Model

- 😊 Can incorporate distant nodes
- 😊 Can mix neighbors across distances
i.e. can learn Gabor Filters!



- 😞 Appendix Experiments of [7] shows no gains beyond 2 layers
- 😞 cannot arbitrary mix neighbors from various distances
→ **cannot learn Gabor Filters!**

Vanilla GCN Model

Addressing Pitfall #2: Scalability

Two approaches:

1- Sampling-based

2- Decoupling-based (aka linearized models)

Addressing Scalability via Sampling

Rather than processing graph as a whole, at each training step, sample a few nodes and edges.

- [uniform] sample training nodes **uniformly at random**, its edges **uniformly at random**
- [biased] e.g., sample high degree nodes with higher probability.
- [learnable] sample from trainable distribution.

Papers: “GraphSAGE”, “**GTTF**”, “TF-GNN”, ...

Papers “FastGCN”, “LADIES”, “ClusterGCN”, “GraphSAINT”, ...

Papers: “DSKReG”, “Performance-adaptive sampling GNN”, “**SubMix**”

GRAPH TRAVERSAL WITH TENSOR FUNCTIONALS: A META-ALGORITHM FOR SCALABLE LEARNING

- Graph Learning algorithms require the graph.
- They out to be **approximable** by graph traversals

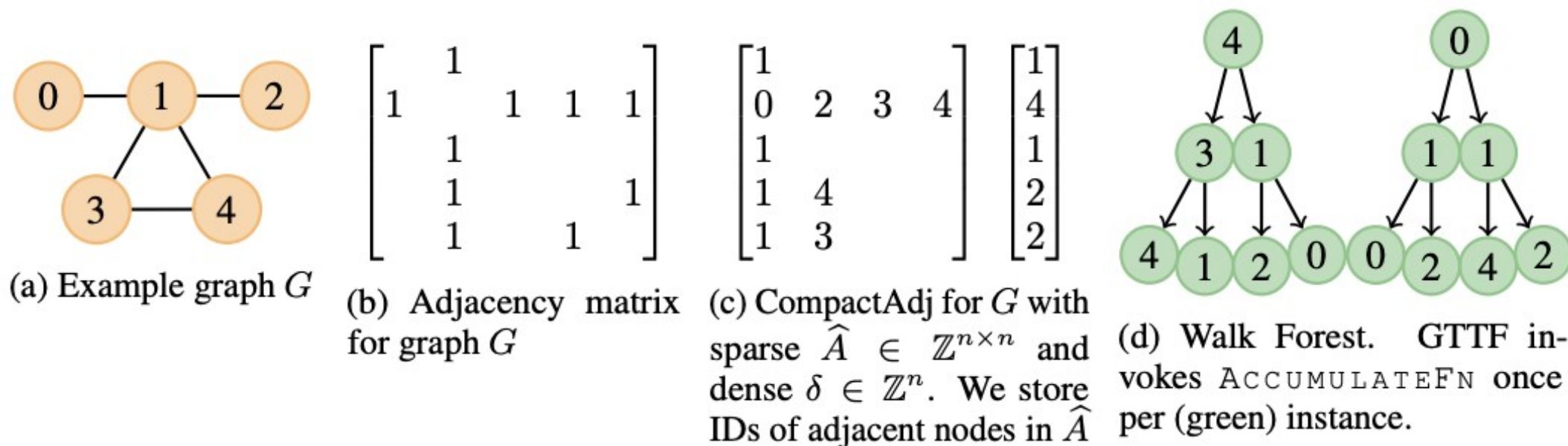


Figure 1: (c)&(d) Depict our data structure & traversal algorithm on a toy graph in (a)&(b).

WEDNESDAY: No class – I'll do
an in-person office hour instead
I'll send an email reminder

